



MARWADI UNIVERSITY

Faculty of **Engineering/ Technology****Information and Communication Technology**

SEM: 5

MU ESE EXAM(Open Book)

B.TECH**DEC: 2024**Subject: - **(DSIP) (01CT1513)**

Date:- 23-12-2024

Total Marks:-50

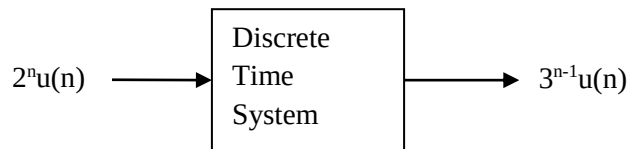
Time: -120 min

Instructions:

1. All Questions are Compulsory.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Do not write/sign/indication/tick mark anything other than Enroll No. at a specific place on the question paper.

Question: 1. CO1, CO2

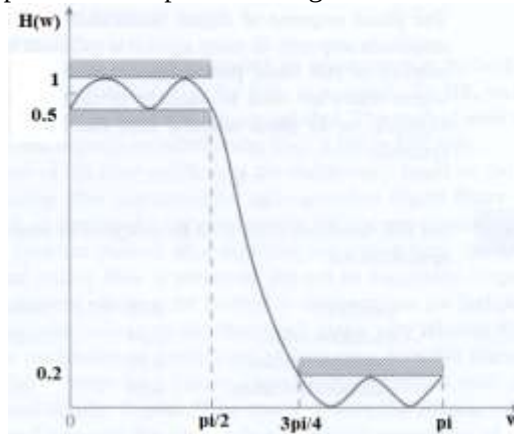
If Discrete time system is defined as following



- (i) impulse response (ii) Unit Step Response (iii) Frequency Response (iv) Magnitude Response (v) Phase Response. [10]

Question: 2. CO5

Design Analog Butterworth low pass Filter as per following characteristics



Determine System Function of Analog Filter and convert it into Digital IIR Filter. Design the structure using Direct form I. Convert following Filter into FIR Filter using rectangular window function. [10]

Question 3. CO3

Draw the histogram of following Image [10]

1 2 5 6 9

2 3 8 5 9

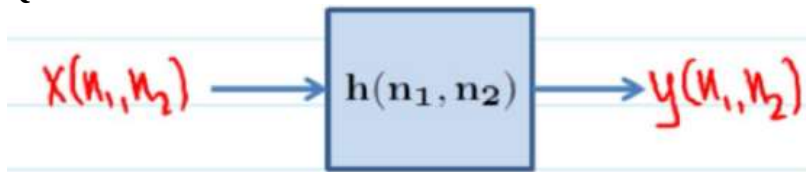
6 1 4 8 0

8 1 0 3 2

Perform the histogram equalization and also draw the equalized histogram.

Perform Image transformation using transfer function

- (i) Output Pixel = 3 * Input Pixel - 5
- (ii) Draw the image negation matrix.

Question 4. CO4**[10]**

$X(n_1, n_2) =$

6	2	3	5	8
9	2	6	5	8
8	6	3	5	7
1	3	2	8	5

$h(n_1, n_2) =$

-1	3	-1
-1	3	-1
-1	4	-1

Find $y(n_1, n_2)$. For image $X(n_1, n_2)$, find the filtered image output using mean filtering, mode filtering and median filtering if structuring element size is 3×3 .

Question 5. CO4**[10]**

If original image is

1	0	1
0	1	0
1	1	1

and structural image is

1	0
0	1

Perform various morphological operations Dilation, Erosion, opening, closing.